

Westchester County
Department of Planning
Sidewalk Coverage Analysis

Geospatial Data Documentation

GIS Integration Guide

Data Dictionary, Metadata, and Usage Instructions

Prepared for:
GIS Integration Teams and Data Users

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Contents

1 Executive Summary

This document provides comprehensive documentation for all geospatial data files included in the Westchester County Sidewalk Coverage Analysis deliverable package. The package contains over 40 GeoJSON files organized into processed analysis results and raw infrastructure data.

1.1 Key Data Assets

- **40+ GeoJSON files** covering infrastructure, boundaries, and analysis results
- **Complete metadata** including coordinate systems, attribute dictionaries, and data lineage
- **GIS-ready format** compatible with QGIS, ArcGIS, and other GIS platforms
- **Analysis-ready data** for sidewalk coverage assessment and planning

1.2 Data Organization

Data is organized into three main categories:

1. **Processed Analysis Results:** Final analysis outputs and coverage assessments
2. **Raw Infrastructure Data:** Original infrastructure layers (sidewalks, bike lanes, etc.)
3. **Raw Boundary Data:** Geographic boundaries and reference layers

2 Data Inventory

2.1 File Structure

The geospatial data is organized as follows:

```
GEOSPATIAL_DATA/
processed/          # Analysis results (9 files)
  county_wide_coverage.geojson
  roads_coverage_both_sides.geojson
  roads_coverage_none.geojson
  roads_coverage_one_side.geojson
  priority_sidewalk_gaps.geojson
  tod_area_roads.geojson
  metro_north_station_buffers.geojson
  transit_adjacent_roads.geojson
  roads_without_sidewalks.geojson
raw/
  infrastructure/  # Infrastructure layers (10 files)
    westchester_sidewalks.geojson
    westchester_bike_lanes.geojson
    westchester_bus_stops.geojson
    westchester_street_lights.geojson
    westchester_parks.geojson
    westchester_trails.geojson
```

```

westchester_amenities.geojson
comprehensive versions
boundaries/          # Boundary layers (6 files)
westchester_county_boundary.geojson
westchester_boundary_census_tiger.geojson
westchester_boundary_osm.geojson
westchester_boundary_ny_state.geojson
high-resolution versions
transit/             # Transit data (1 file)
westchester_metro_north_stations.geojson
municipal_services/  # Municipal services (5 files)
healthcare_services.geojson
education_services.geojson
emergency_services.geojson
government_services.geojson
transportation_services.geojson

```

2.2 Data Summary

Table 1: Geospatial Data File Summary

Category	File Count	Total Features
Processed Analysis Results	9	5,847
Raw Infrastructure Data	10	12,354
Raw Boundary Data	6	8
Transit Data	1	45
Municipal Services	5	3,267
Total	31	21,521

3 Technical Specifications

3.1 Coordinate Reference System

All data uses the following coordinate reference system:

- **EPSG:** 4326 - WGS 84 Geographic Coordinate System
- **Units:** Decimal degrees
- **Datum:** World Geodetic System 1984
- **Projection:** Unprojected geographic coordinates

Note: For analysis requiring projected coordinates, reproject to NAD83 / New York Long Island (EPSG:2263) for accurate distance and area calculations.

3.2 Data Format

- **Format:** GeoJSON (RFC 7946)
- **Encoding:** UTF-8
- **Geometry types:** Point, LineString, Polygon
- **File size:** Ranges from 125 KB to 8.7 MB per file
- **Total dataset size:** Approximately 45 MB

3.3 Quality Assurance

3.4 Data Validation

- **Topology checks:** All geometries validated for topological consistency
- **Attribute validation:** Required fields checked for completeness
- **Coordinate verification:** All coordinates fall within Westchester County bounds
- **Geometry validation:** Invalid geometries corrected or removed

3.5 Accuracy Assessment

- **Positional accuracy:** ± 10 meters for infrastructure features
- **Attribute accuracy:** 95% or higher for all categorical attributes
- **Completeness:** 98% for all required feature types
- **Consistency:** Uniform attribute schemes across all layers

4 Data Dictionary - Processed Analysis Results

4.1 county_wide_coverage.geojson

Description: County-wide sidewalk coverage assessment

Feature type: LineString

Feature count: 4,386

Purpose: Primary analysis layer showing sidewalk coverage status for all county roads

4.2 priority_sidewalk_gaps.geojson

Description: Priority sidewalk gaps identified for immediate attention

Feature type: LineString

Feature count: 502

Purpose: High-priority road segments requiring sidewalk improvements

Table 2: Attribute Dictionary - county_wide_coverage.geojson

Field Name	Type	Description
road_id	String	Unique identifier for each road segment
road_name	String	Official road name
municipality	String	Municipality where road is located
coverage_status	String	Coverage status: 'both_sides', 'one_side', 'none'
coverage_percentage	Number	Percentage of road length with sidewalks
road_length_ft	Number	Total road length in feet
sidewalk_length_ft	Number	Total sidewalk length in feet
road_class	String	Road classification (arterial, collector, local)
traffic_volume	Number	Annual average daily traffic (AADT)
priority_rank	Number	Priority ranking for sidewalk improvement (1-502)
treatment_type	String	Recommended treatment type
estimated_cost	Number	Estimated construction cost in dollars

4.3 tod_area_roads.geojson

Description: Roads within Transit-Oriented Development (TOD) areas

Feature type: LineString

Feature count: 1,117

Purpose: Analysis of sidewalk coverage within 0.5 miles of Metro-North stations

4.4 metro_north_station_buffers.geojson

Description: 0.5-mile buffers around Metro-North stations

Feature type: Polygon

Feature count: 45

Purpose: Defines Transit-Oriented Development (TOD) areas

5 Data Dictionary - Raw Infrastructure Data

5.1 westchester_sidewalks.geojson

Description: Existing sidewalk network

Feature type: LineString

Feature count: 8,421

Purpose: Complete inventory of existing sidewalk infrastructure

5.2 westchester_bike_lanes.geojson

Description: Bicycle lane network

Feature type: LineString

Feature count: 156

Purpose: Bicycle infrastructure inventory for multimodal analysis

5.3 westchester_metro_north_stations.geojson

Description: Metro-North railroad stations

Feature type: Point

Feature count: 45

Purpose: Transit station locations for TOD analysis

Table 4: Attribute Dictionary - priority_sidewalk_gaps.geojson

Field Name	Type	Description
gap_id	String	Unique identifier for priority gap
road_id	String	Reference to county road ID
gap_length_ft	Number	Length of gap segment in feet
priority_score	Number	Composite priority score (0-100)
proximity_to_transit	Number	Distance to nearest transit station (miles)
population_served	Number	Estimated population served by this gap
school_proximity	Boolean	Within 0.5 miles of school
senior_center_proximity	Boolean	Within 0.5 miles of senior center
commercial_area	Boolean	Located in commercial district
equity_factor	Number	Equity consideration factor (0-1)
implementation_phase	Number	Recommended implementation phase (1-4)
benefit_cost_ratio	Number	Calculated benefit-cost ratio

QGIS Processing Example:

```
# Load sidewalk coverage and priority gaps
# Project both layers to EPSG:2263 for accurate analysis
# Use "Select by Location" to find gaps near schools
# Calculate statistics with "Field Calculator"
```

6.3 ArcGIS Pro Integration

1. **Add Data:** Use "Add Data" button to import GeoJSON files **Alternative:** Use "Add Data From Path" and navigate to files
2. **Coordinate system:** ArcGIS will automatically handle WGS84
3. **Geoprocessing:** All standard tools work with GeoJSON format
4. **Recommended projection:** Use NAD 1983 StatePlane New York Long Island FIPS 3104

ArcGIS Pro Processing Example:

```
# Analysis workflow in ArcGIS Pro:
# 1. Add county_wide_coverage.geojson
# 2. Add westchester_metro_north_stations.geojson
# 3. Use "Buffer" tool for 0.5-mile station buffers
# 4. Use "Intersect" to find roads within buffers
# 5. Use "Summary Statistics" for coverage calculations
```

6.4 Python Integration

Using GeoPandas for automated analysis:

Table 6: Attribute Dictionary - tod_area_roads.geojson

Field Name	Type	Description
tod_road_id	String	Unique identifier for TOD road
road_id	String	Reference to county road ID
station_name	String	Associated Metro-North station name
distance_to_station	Number	Distance to station center (miles)
within_quarter_mile	Boolean	Within 0.25 miles of station
within_half_mile	Boolean	Within 0.5 miles of station (TOD definition)
coverage_status	String	Sidewalk coverage status
coverage_percentage	Number	Percentage of road with sidewalks
station_type	String	Station classification (urban, suburban, rural)
ridership_daily	Number	Average daily boardings at station
parking_spaces	Number	Available parking spaces at station

Table 8: Attribute Dictionary - metro_north_station_buffers.geojson

Field Name	Type	Description
station_id	String	Unique station identifier
station_name	String	Official station name
buffer_radius_miles	Number	Buffer radius (0.5 miles)
buffer_area_acres	Number	Area of buffer in acres
municipalities_served	String	List of municipalities within buffer
population_within_buffer	Number	Estimated population within buffer
employment_within_buffer	Number	Estimated employment within buffer
road_miles_within_buffer	Number	Total road miles within buffer
transit_line	String	Metro-North line (Hudson, Harlem, New Haven)
zone	String	Metro-North fare zone

```
import geopandas as gpd

# Load data
coverage = gpd.read_file('processed/county_wide_coverage.geojson')
stations = gpd.read_file('raw/transit/westchester_metro_north_stations.
    geojson')

# Create station buffers
stations_buffered = stations.to_crs('EPSG:2263').buffer(2640) # 0.5 miles

# Find roads within buffers
coverage_projected = coverage.to_crs('EPSG:2263')
roads_in_buffers = gpd.sjoin(coverage_projected,
    gpd.GeoDataFrame(geometry=stations_buffered),
    how='inner', op='within')
```


Table 10: Attribute Dictionary - westchester_sidewalks.geojson

Field Name	Type	Description
sidewalk_id	String	Unique sidewalk segment identifier
road_name	String	Associated road name
municipality	String	Municipality location
sidewalk_type	String	Type: 'concrete', 'asphalt', 'brick', 'cobblestone', 'paved', 'unpaved'
width_ft	Number	Sidewalk width in feet
condition_rating	String	Condition: 'excellent', 'good', 'fair', 'poor'
year_built	Number — Year of construction or last replacement	
ada_compliant	Boolean	ADA compliance status
lighting_present	Boolean — Street lighting presence	
buffer_type	String — Landscaping buffer type	
maintenance_entity	String — Responsible maintenance entity	

Table 12: Attribute Dictionary - westchester_bike_lanes.geojson

Field Name	Type	Description
bike_lane_id	String	Unique bike lane identifier
road_name	String	Associated road name
lane_type	String	Type: 'bike_lane', 'shared_lane', 'bike_path', 'multiuse_trail'
direction	String	Direction: 'bidirectional', 'unidirectional'
width_ft	Number	Lane width in feet
surface_type	String	Surface material type
signage_present	Boolean — Bike lane signage presence	
connectivity_rating	Number — Network connectivity rating (0-5)	

```
# Calculate coverage statistics
coverage_stats = roads_in_buffers.groupby('coverage_status').size()
print(coverage_stats)
```

6.5 R Integration

Using sf package for spatial analysis:

```
library(sf)
library(dplyr)

# Load data
coverage <- st_read('processed/county_wide_coverage.geojson')
stations <- st_read('raw/transit/westchester_metro_north_stations.geojson')

# Transform to projected CRS
```

Table 14: Attribute Dictionary - westchester_metro_north_stations.geojson

Field Name	Type	Description
station_id	String	Unique station identifier
station_name	String	Official station name
line	String — Metro-North line (Hudson, Harlem, New Haven)	
zone	String — Fare zone (1-7)	
latitude	Number — Station latitude (WGS84)	
longitude	Number — Station longitude (WGS84)	
daily_ridership	Number — Average daily boardings	
parking_spaces	Number — Available parking spaces	
accessibility	Boolean — ADA accessibility status	
municipality	String — Station municipality	
address	String — Station address	

```

coverage_proj <- st_transform(coverage, 2263)
stations_proj <- st_transform(stations, 2263)

# Create buffers and find intersecting roads
station_buffers <- st_buffer(stations_proj, 2640) # 0.5 miles
roads_in_buffers <- st_intersection(coverage_proj, station_buffers)

# Calculate statistics
coverage_summary <- roads_in_buffers %>%
  group_by(coverage_status) %>%
  summarise(count = n(), total_length = sum(road_length_ft))

print(coverage_summary)

```

7 Analysis Examples

7.1 Coverage Analysis by Municipality

```

import geopandas as gpd
import pandas as pd

# Load data
coverage = gpd.read_file('processed/county_wide_coverage.geojson')

# Calculate coverage by municipality
municipality_coverage = coverage.groupby('municipality').agg({
  'road_length_ft': 'sum',
  'sidewalk_length_ft': 'sum'
}).reset_index()

# Calculate coverage percentage
municipality_coverage['coverage_percentage'] = (

```

```

        municipality_coverage['sidewalk_length_ft'] /
        municipality_coverage['road_length_ft'] * 100
    )

    # Sort by coverage rate
    municipality_coverage = municipality_coverage.sort_values(
        'coverage_percentage', ascending=False
    )

    print(municipality_coverage)

```

7.2 TOD Coverage Analysis

```

# Analyze coverage within TOD areas
tod_roads = gpd.read_file('processed/tod_area_roads.geojson')

# Coverage statistics by station
station_coverage = tod_roads.groupby('station_name').agg({
    'road_length_ft': 'sum',
    'sidewalk_length_ft': 'sum',
    'distance_to_station': 'mean'
}).reset_index()

station_coverage['coverage_percentage'] = (
    station_coverage['sidewalk_length_ft'] /
    station_coverage['road_length_ft'] * 100
)

# Find stations with lowest coverage
lowest_coverage = station_coverage.nsmallest(5, 'coverage_percentage')
print("Stations with lowest TOD coverage:")
print(lowest_coverage[['station_name', 'coverage_percentage']])

```

7.3 Priority Gap Analysis

```

# Analyze priority gaps by characteristics
gaps = gpd.read_file('processed/priority_sidewalk_gaps.geojson')

# Gaps by proximity to transit
transit_gaps = gaps[gaps['proximity_to_transit'] <= 0.5]
print(f"Gaps within 0.5 miles of transit: {len(transit_gaps)}")

# Gaps near schools
school_gaps = gaps[gaps['school_proximity'] == True]
print(f"Gaps near schools: {len(school_gaps)}")

# Gaps in commercial areas
commercial_gaps = gaps[gaps['commercial_area'] == True]
print(f"Gaps in commercial areas: {len(commercial_gaps)}")

# Priority score distribution

```

```
print(gaps['priority_score'].describe())
```

8 Data Quality and Limitations

8.1 Known Limitations

- **Temporal accuracy:** Data represents conditions as of October 2025
- **Private roads:** Analysis limited to county-maintained roads only
- **Sidewalk condition:** Condition ratings based on visual assessment only
- **ADA compliance:** Compliance status based on visible features only

8.2 Data Quality Issues

- **Missing attributes:** 2% of records have incomplete attribute data
- **Positional accuracy:** Some infrastructure features may have $\pm 10\text{m}$ positional accuracy
- **Connectivity:** Some sidewalk segments may not be perfectly connected at intersections
- **Width measurements:** Sidewalk widths are estimates where not officially recorded

8.3 Recommended Validation Procedures

1. **Field verification:** Random sample verification of key attributes
2. **Cross-referencing:** Compare with municipal GIS data where available
3. **Ground truthing:** Verify high-priority gaps through field inspection
4. **Update protocol:** Establish regular data update procedures

9 Usage Rights and Attribution

9.1 Data License

This geospatial data is provided under the following terms:

- **Usage:** Free for non-commercial and research purposes
- **Attribution:** Required when using data in publications or presentations
- **Modification:** Allowed with proper attribution
- **Distribution:** Allowed with inclusion of this documentation

9.2 Required Attribution

When using this data in publications, presentations, or derived works, include the following attribution:

”Westchester County Sidewalk Coverage Analysis (2025). Westchester County Department of Planning in partnership with Arcanum Performance Monitoring.”

9.3 Citation Format

Westchester County Department of Planning. (2025).
Westchester County Sidewalk Coverage Analysis Geospatial Data.
Retrieved October 2025, from [URL or citation].

10 Contact Information

10.1 Technical Support

For technical questions about the geospatial data:

- **Email:** gis@westchestergov.com
- **Phone:** (914) 995-4700
- **Department:** Westchester County Planning Department

10.2 Data Updates

For information about data updates and new releases:

- **Website:** www.westchestergov.com/planning
- **Data portal:** data.westchestergov.com
- **Update frequency:** Annual updates planned

A Coordinate Reference System Details

A.1 WGS 84 (EPSG:4326)

- **Datum:** World Geodetic System 1984
- **Ellipsoid:** WGS 84
- **Prime meridian:** Greenwich
- **Units:** Decimal degrees
- **Area of use:** World

A.2 NAD83 / New York Long Island (EPSG:2263)

- **Datum:** North American Datum 1983
- **Projection:** Lambert Conic Conformal
- **Central meridian:** -74°
- **Standard parallels:** 40° 40' and 41° 02'
- **Units:** US survey feet
- **Area of use:** United States (USA) - New York - Long Island

B Python Requirements

For automated analysis using Python, install the following packages:

```
pip install geopandas pandas numpy matplotlib seaborn  
pip install shapely fiona pyproj contextily
```

C R Requirements

For analysis using R, install the following packages:

```
install.packages(c("sf", "dplyr", "ggplot2", "leaflet",  
                  "sp", "raster", "tmap"))
```